



**Students'
Academic
Cell**



BITS Pilani
Pilani | Dubai | Goa | Hyderabad

Minor Programmes for First Degree Students at BITS

Compiled by :
Students' Academic Cell
BITS Pilani

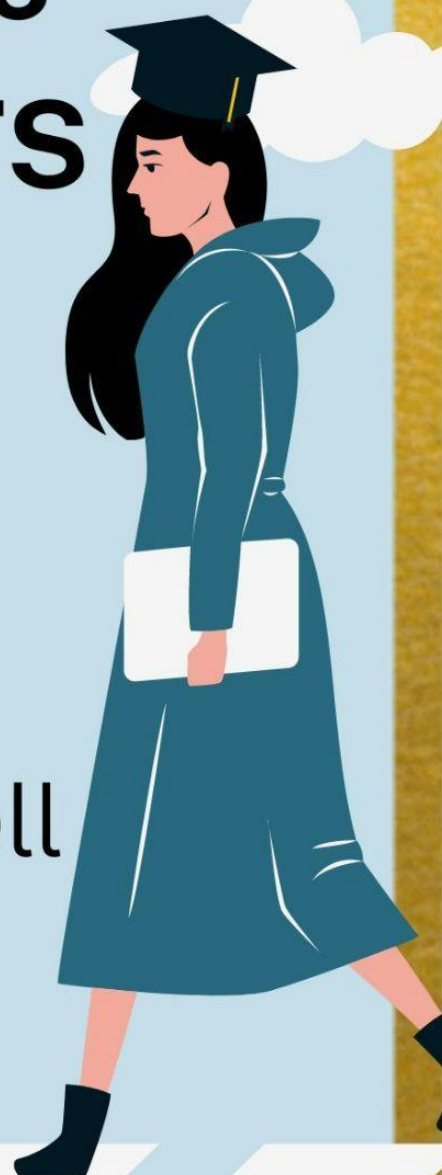


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Introduction

"Minor programmes" are offered as choices for first-degree students with the purpose of encouraging them to add attention to their supplemental learning (outside of a major subject), as well as recognising and certifying the information earned in an area that is outside of their major.

At the end of the second academic year, a student must decide whether he/she would prefer to pursue a minor and ideally begin the application process for the same.

At present Sixteen minor programs viz. **Minor in Aeronautics, Computational Economics, Computing and Intelligence, Data Science, English Studies, Entrepreneurship, Film and Media, Finance, Management, Materials Science and Engineering, Philosophy, Economics and Politics (PEP), Physics, Public Policy, Robotics and Automation, Supply Chain Analytics and Water and Sanitation** have been designed which are selectively being offered at different campuses.

This document contains valuable insights and information that one can refer to for an idea about all the minor programmes offered by BITS.

Minor fee

- If any continuing student is also admitted to a minor programme, he/she has to pay a fee of Rs. 26,000/- for AY 2022-23 in addition to fees for the semesters / summer terms enrolled in. The fee is payable in two instalments – Rs. 13,000/- at the time of admission as minor application fee and Rs. 13,000/- on completion of requirements as minor certificate fee. Admission fees for minor programme will be revised in subsequent years as per Institute norms.
- Minor Certificate Fee is 6% of semester tuition fee in corresponding Academic Year.
- If any continuing international student is also admitted to a minor programme, then he/she has to pay the minor fee and in two instalments. The first instalment of the fee will be paid at the start of the minor program and it will be 3% of semester tuition fee in the corresponding academic year. The second instalment will be paid at the completion of minor program and it will be 3% of tuition fee in the semester when he /she complete minor program. This fee will be in addition to the semester/term fees. Admission fees for minor programme will be revised in subsequent years as per Institute norms.

Requirements for a minor

- **Courses and unit requirements:**

Each minor would be defined by coursework requirement with the following conditions:

Category	Courses	Units
Minor – Core	4 (max)	12 (max)
Minor – Electives	2 (min)	6 (min)
Minor – Total	5 (min)	15 (min)

- **Elective Pool:**

The pool of electives specific to a minor may include courses from one or more disciplines and may include project / seminar type courses.

A student may use at most one project / seminar type course to meet the requirements of a minor.

- **Overlap in requirements:**

At most 2 courses (and at most 6 units) out of the above requirement (of 5 courses and 15 units) may be met by mandatory courses of the student's degree i.e., major (or degrees i.e., majors): i.e., from the general institutional requirement (excluding Humanities requirement) or the (Major) discipline Core(s).

No course may be used to meet the requirements of two different minors nor may a course be used to the meet the requirements of two majors and a minor.

- **GPA requirement:**

A student – on completion of the requirements for a minor – must have maintained a cumulative GPA of 4.5 or above (out of 10) in the courses applied to the minor.

- **Process for declaring / obtaining a minor**

1. A student – if he/she chooses to pursue a minor – must declare at the end of the 2nd year that he/she will pursue a specific minor.
2. The student will be charged a small fee for logistics.
3. If and when he/she completes the requirements for the minor – as stipulated above and as stipulated for the specific minor, then he/she may apply for a “minor” certificate.
4. If it is verified that the requirements are met then he/she will be awarded a “minor certificate” (separate from a degree – i.e. major – certificate).
5. A minor certificate will be issued only on completion of a degree (i.e. a major).

Interview with Mr. AP Singh, Dean

AUGSD regarding Minors:

Out of the 16 minors offered, 4 have been recently introduced- Supply Chain Analytics, Computing and Intelligence, Computational Economics and Management. Mr. AP Singh mentioned that the students should be made more aware regarding the minors offered at Pilani campus.

Computing and Intelligence is one of the new minors offered at the Pilani campus, however it was established a few years ago at the Hyderabad campus, where 212 students enrolled. In Computational Economics, there were more than twenty students. Every new academic minor or decision applies equally to all three campuses.

The institute takes care while establishing academic programmes so that students do not have to do additional work. The courses and prerequisites are flexible and modular.

Module-1: General Institutional Requirements foundation courses- Aiming at professional development, courses on general knowledge such as professional ethics, POE, POM, and TRW. BITS is distinctive with regard to general institution requirements.

Core/ Departmental Electives - Module-2

HuEls - Module-3: over 50-75 courses

OpEls - Module - 4: another BITS-exclusive feature

He advised the students to plan for prerequisites when pursuing a minor degree.

The courses of a minor degree can be completed as OpEls, but a minor degree is more structured and demonstrates a degree of specialty.

A minor enhances your profile while applying for placements. One can also prepare for the civil service by completing a minor in Public Policy or Philosophy, Economics, and Politics. Mr. AP Singh mentioned that Minor in Water and Sanitation is one of the least popular minors.

Several institutes funded by the government of India and the United Nations offer various incentives to promote research and development in a subject that is directly helpful to society. Institutes such as Ahmedabad Global Sanitation Graduate School are eager for additional graduates in this discipline and are willing to offer substantial financial aid for this purpose. Today's sectors require people with particular skills.

Minor programmes are designed to be inter-disciplinary in character, one must decide for oneself whether the minor is the best fit. He/she should not rely too heavily on the recommendations of others. Each combination of a minor and a major degree is viable if the student can identify the optimum application of his minor in his major field. One may also choose a minor out of pure interest.

The Bioscience area has recently witnessed a great deal of innovation, and the knowledge gained through the Minor in Automation and Robotics can be utilised in novel ways in the field.

In the 21-22 academic year at the Pilani campus, 57% of the overall batch was placed in the IT industry, of whom just 18% were dedicated CS students.

This demonstrates that the remaining candidates exploited their minors or electives to persuade the recruiter that they were on par with students majoring in the same field.

One minor should not be more popular than the others, and it should not be pigeonholed without knowledge of its merits. 270 out of more than 300 applicants for a Minor in Finance were chosen. CG criteria (cut-offs) are typically in place to avoid overcrowding a department so that teachers are also comfortable when instructing. A minor is only a good option if the student is genuinely interested in pursuing it.

Dual degree students already have a strong profile due to their two degrees. Typically, dual degree students finish their minor course requirements during the summer term. In addition to this, caution must be given when planning minor courses for the summer term, as not all courses are available. If one decides to pursue a minor without finishing a summer term, he/she may do so by completing the remaining courses in 5-1 or 5-2 (the semester without a PS in year 5), but he/she must choose a shorter thesis and may take no more than three more courses. This further restricts his or her double PS option.

Across all Indian campuses, the average PS-2 stipend was INR 60,000. If a student desires a minor degree that is not currently being provided by the institution due to low enrolment, he may speak with the head of the relevant department. In the past, several programmes were offered even if there were a very small number of students.

A minor in English Studies would be useful for applying overseas for a Master of Science degree, as well as for TOEFL, GRE, and language certification.

A minor in Public Policy, PEP, offers general education courses and assists in the preparation for Civil Service Exams. At the interview

level, 300 out of 400 UPSC applicants were from the Civil sector. If students have a firm grasp of economics concepts, they may find civil examinations to be less challenging than their peers.

Minors offered at BITS, Pilani Campus

1. Minor in Aeronautics

Courses & Units Req.	06 courses (min) 18 units (min)				
Core Courses	Course Number	Course Title	L	P	U
	AN F311	Principles of Aerodynamics	3	0	3
	AN F312	Aircraft Propulsion	3	0	3
	AN F313	Flight Mechanics and Controls	3	0	3
Electives	AN F314	Introduction to Flight	3	0	3
	AN F315	Aircraft Structures	3	0	3
	ME F415	Gas Dynamics	3	0	3
	ME F418	Rocket and Spacecraft Propulsion	3	0	3
	ME F452	Composite Materials and Design	3	0	3
	ME F482	Combustion	3	0	3
	ME F485	Numerical Techniques for Fluid Flow & Heat Transfer	3	0	3
	EEE F242	Control Systems	3	0	3
	EEE F417	Computer Based Control Systems	3	0	3
	ME F376	Design Project			3

Spanning 3 core courses and 10 electives from which a minimum of 18 units must be chosen, Aeronautics is an intriguing field that encompasses the principles of aerodynamics (the interaction of air with moving things), propulsion (power systems responsible for the generation of thrust for supplying motion), and structures (the construction of structures).

It also involves flight mechanics (trajectory analysis and material properties), airframes, and applying optimization to airborne vehicles operating within the Earth's atmosphere, and to the outer rockets and spacecraft.

2. Minor in Computational Economics

Courses & Units Req.	05 courses (min) 15 units (min)				
Core Courses	Course Number	Course Title	L	P	U
	ECON F215	Computational Methods for Economics	3	0	3
	ECON F241	Econometric Methods	3	0	3
	ECON F242	Microeconomics	3	0	3
Electives	BITS F314	Game Theory and its Applications	3	0	3
	BITS F464	Machine Learning	3	0	3
	CS F320	Foundations of Data Science	3	0	3
	ECON F342	Applied Econometrics	3	0	3
	ECON F419	Advanced Microeconomics	3	0	3
	ECON F420	Applied Macroeconometrics	3	0	3
	MATH F424	Applied Stochastic Process	3	1	4

This minor consists of 3 core courses and 7 electives from which a minimum of 15 units must be selected to complete the degree.

The emergence of the unified discipline of economics, mathematics, and computer science resulted from the convergence of intellectual requirements for interdisciplinary education and research. The modern instruments and methods utilized by computer scientists have evolved.

Students with an interest in computational mechanism design with applications to economics and especially for those with a broader interest would benefit from this minor.

Data is necessary for making effective economic judgments. The schedule integrates the educational talents of multiple departments and important computational skills related to economics and job preparation such as economics, finance, and business operations.

Considering this robust interdisciplinarity, it is also a notable addition to being a good preparation for graduate study in the field of economics or decision sciences.

3. Minor in Computing and Intelligence

Courses & Units Required	06 courses (min) 18 units (min)				
Core Courses	Course Number	Course Title	L	P	U
	BITS F232	Foundations of Data Structures and Algorithms	3	1	4
	CS F372	Operating Systems	3	0	3
	CS F407	Artificial intelligence	3	0	3
Electives	BITS F311	Image Processing	3	0	3
	BITS F452	Blockchain Technology	3	0	3
	BITS F463	Cryptography	3	0	3
	BITS F464	Machine Learning	3	0	3
	CS F212	Database Systems	3	1	4
	CS F213	Object Oriented Programming	3	1	4
	CS F301	Principles of Programming Languages	2	0	2
	CS F303	Computer Networks	3	1	4
	CS F314	Software Development for Portable Devices	2	1	3
	CS F315	Information and Communication Technologies and Development	3	0	3
	CS F415	Data Mining	3	0	3
	IS F311	Computer Graphics	3	0	3
	IS F341	Software Engineering	3	1	4

Comprising 3 core courses and 13 electives, Computing and Intelligence is a rather popular choice among the students.

The Minor in Computing and Intelligence enables non-Computer Science majors to gain a deeper understanding of computing and artificial intelligence and apply this knowledge to problem-solving in diverse domains, while courses such as Data Structures, Foundations and Algorithms would aid students in abstract reasoning and problem solving.

This minor programme is exclusive to first-degree students of disciplines other than Computer Science.

4. Minor in Data Science

Courses & Units Required	5 courses (min) 15 units (min)				
Core Courses	Course Number	Course Title	L	P	U
	BITS F464	Machine Learning	3	0	3
	CS F320	Foundations of Data Science	3	0	3
	MATH F432	Applied statistical Methods	3	0	3
Electives	BITS F453	Computational Learning Theory	3	0	3
	BITS F454	Bio-Inspired Intelligence: Algorithms and Applications	3	0	3
	CS F317	Reinforcement Learning	3	0	3
	CS F407	Artificial Intelligence	3	0	3
	CS F415	Data Mining	3	0	3
	CS F425	Deep Learning	3	0	3
	CS F426	Graph Mining	3	1	4
	CS F429	Natural Language Processing	3	0	3
	CS F432	Brain-inspired Deep Learning	3	0	3
	CS F433	Computational Neuroscience	3	0	3
	CS F469	Information Retrieval	3	0	3
	CS G519	Social Media Analytics	3	1	4
	MATH F212	Optimization	3	0	3
	MATH F353	Statistical Inference and applications	3	0	3
	MATH F424	Applied Stochastic Processes	3	1	4
	MATH F471	Nonlinear Optimization	3	0	3

This minor is also among the frequently chosen minors, consisting of 3 core courses and 14 electives from which a minimum of 15 units must be taken.

The minor in Data Science is designed to teach students the fundamental skills required of Data Scientists in the modern world.

Data Science is becoming pervasive across all industries, creating new business opportunities. The minor will assist students in applying Mathematics, Statistics, and Computer Science for analysing information gathered from various sources in their respective engineering applications and providing insights that are meaningful and actionable.

5. Minor in English Studies

Courses & Units Required	5 courses (min) 15 units (min)				
Core Courses	Course Number	Course Title	L	P	U
	GS F241	Creative Writing	2	1	3
	HSS F337	English Literary Forms and Movements	3	0	3
Electives Pool – I (Language)	GS F221	Business Communication	3	0	3
	GS F244	Reporting and Writing for Media	3	0	3
	GS F245	Effective Public Speaking	3	0	3
	HSS F222	Linguistics	3	0	3
	HSS F227	Cross Cultural Skills	3	0	3
	HSS F228	Phonetics and Spoken English	3	0	3
	HSS F342	Advanced Communicative English	3	0	3
Elective Pool-II (Literature)	GS F242	Cultural Studies	3	0	3
	GS F322	Critical Analysis of Literature and Cinema	3	0	3
	HSS F221	Readings from Drama	3	0	3
	HSS F226	Postmodernism	3	0	3
	HSS F237	Contemporary Indian English Fiction	3	0	3
	HSS F316	Popular Literature and Culture of South Asia	3	0	3
	HSS F327	Contemporary Drama	3	0	3
	HSS F330	Appreciation of Art	3	0	3
	HSS F332	Cinematic Arts	3	0	3
	HSS F335	Literary Criticism	3	0	3
	HSS F336	Modern Fiction	3	0	3
	HSS F338	Comparative Indian Literature	3	0	3
	HSS F340	Postcolonial Literatures	3	0	3
	HSS F349	Ecocriticism	3	0	3
	HSS F373	Shakespeare and Popular Culture	3	0	3
	HSS F399	Introduction to American Literature	3	0	3

Offered by the department of Humanities and Social Sciences, English Studies comprises 2 core courses and an elective pool 1 (Language) of 7 electives and pool 2 (Literature) of 16 electives.

Language, literature, and culture are abundant in English. Classic English literature is still read and admired. English has likewise evolved and is now the dominant language in communication in

business, diplomacy, media, global education, etc. The Minor in English Studies introduces students to the language and literary canons and gives them sufficient exposure not just to cultural and linguistic aspects but also practical applications of Literature and English.

In particular, core and optional courses in the Minor would stimulate literary criticism and linguistic analyses, and the ability to meaningfully analyse interpretation and explanation.

6. Minor in Entrepreneurship

Courses & Units Required	5 courses (min) 15 units (min)				
Core Courses	Course Number	Course Title	L	P	U
	BITS F468	New Venture Creation	3	0	3
	BITS F482 or ECON F414	Creating and Leading Entrepreneurial Organizations	3	0	3
	ECON F212	Fundamentals of Finance and Accounting	3	0	3
Electives (minimum of 2 courses and additional units required to make the total to 15)	BITS F322	Venture Team Development and Organization	3	0	3
	BITS F323	Venture Finance	3	0	3
	BITS F324	Strategy for Entrepreneurs	3	0	3
	BITS F325	New Product and Service Design	3	0	3
	BITS F326	Design Thinking for Innovation & Entrepreneurship	3	0	3
	BITS F427	Digital Marketing	3	0	3

This minor contains 3 core courses and 6 electives from which 2 courses and additional units are required to make the total to 15.

Entrepreneurship solves market and societal issues through innovation and job creation. It trains students from various fields in the knowledge of entrepreneurship, skills, and experience to create innovative products/services for new entrepreneurs and ventures.

This programme emphasizes practical learning, giving students a chance to learn in teams, design innovative products and start businesses.

7. Minor in Film and Media

Courses & Units Required		6 courses (min) 18 units (min)			
	Course number	Course Title	L	P	U
Core Courses	GS F223	Introduction to Mass Communication	3	0	3
	GS F244	Reporting and Writing for Media	3	0	3
	GS F322	Critical Analysis of Literature and Cinema	3	0	3
Elective Courses	GS F224	Print and Audio Visual Advertising	3	0	3
	GS F242	Cultural Studies	3	0	3
	GS F321	Mass Media Content and Design	3	0	3
	GS F343	Short Film and Video Production	3	0	3
	HSS F332	Cinematic Arts	3	0	3

With 3 core courses and 5 electives, this minor requires 18 units for successful completion.

As the name suggests, it involves films and other types of media that are directly or indirectly derived from it, such as television and advertising, and that they are the most influential cultural influences in the modern world.

The following are some of the goals for the Film and Media minor:

- i. An introductory survey of the field of media studies, with a particular emphasis on film studies
- ii. An introduction to the fundamentals of print and digital media, including video production and film appreciation
- iii. Practical instruction in writing for the media and the production of films.

8. Minor in Finance

Courses & Units Required	5 courses (min) 15 units (min)				
Core Courses	Course Number	Course Title	L	P	U
	ECON F212	Fundamentals of Finance and Accounting	3	0	3
	FIN F315	Financial Management	3	0	3
Elective Courses	ECON F241	Econometric methods	3	0	3
	ECON F312	Money banking and Financial markets	3	0	3
	ECON F355	Business Analysis & Valuation	3	0	3
	ECON F411	Project Appraisal	3	0	3
	ECON F413	Financial Engineering	3	0	3
	FIN F242	Introduction to Financial Mathematics	3	0	3
	FIN F243	Functions & Working of Stock Exchanges	3	0	3
	FIN F311	Derivatives & Risk Management	3	0	3
	FIN F312	Fundamentals of Taxation and Audit	3	0	3
	FIN F313	Security Analysis & Portfolio Management	3	0	3
	FIN F314	Investment Banking & Financial Services	3	0	3
	FIN F414	Financial Risk Analytics and Management	3	0	3

Finance includes 2 core courses and 12 electives, from which 15 units must be taken.

The minor in Finance seeks to provide students with a foundation in accounting and finance so as to supplement their existing disciplinary knowledge, enrich their educational experience, and help them to make more informed decisions.

It will also provide students with a chance to get knowledge about investing and quantitative applications in finance.

9. Minor in Management

Courses & Units Required	05 courses (min) 15 units (min)				
Core Courses	Course Number	Course Title	L	P	U
	BITS F428	Essentials of Strategic Management	3	0	3
	MGTS F211	Principles of Management	3	0	3
	MGTS F314	Essentials of Financial Management	3	0	3
Electives	BITS F326	Design Thinking for Innovation and Entrepreneurship	3	0	3
	ECON F415	New Venture Creation	3	0	3
	ECON F434	International Business	3	0	3
	ECON F435	Marketing Research	3	0	3
	HSS F328	Human Resources Development	3	0	3
	MF F219	Operations Management	3	0	3
	MF F319	Supply Chain Management	3	0	3
	ME F443	Quality Control, Assurance and Reliability	3	0	3
	MGTS F311	Marketing	3	0	3
	MGTS F313	Product and Brand Management	3	0	3
	MGTS F315	Foundations of Business Analytics	3	0	3
	MGTS F316	Managerial and Leadership Skills	3	0	3
	MGTS F351	Organizational Behaviour	3	0	3

Offered by the department of Management, this minor contains 3 core courses and 13 electives designed for the student who seeks a broad introduction to business operations and the development of business acumen.

By understanding the areas of management, the student will be able to possess a competitive edge on the market and throughout their careers and be better prepared to manage projects in practice school through comprehension of managerial and organizational issues. It would also facilitate their technical and managerial skills and aid in investigating the field.

10. Minor in Material Science and Engineering

Courses & Units Required	5 courses (min) 15 units (min)				
Core Courses	Course Number	Course Title	L	P	U
	CHE F243 / ME F213	Materials Science and Engineering	3 2	0 0	3 2
	MST F331	Materials Characterization	3	1	4
	MST F332	Materials Processing	3	0	3
Elective Courses	BITS F416	Introduction to Nanoscience	3	0	3
	CHE F433	Corrosion Engineering	3	0	3
	CHEM F223	Colloid and Surface Chemistry	3	0	3
	CHEM F326	Solid State Chemistry	3	0	3
	CHEM F336	Nanochemistry	3	1	4
	ME F452	Composite Materials and Design	3	0	3
	MST F333	Introduction to Biomaterials	3	0	3
	MST F334	Materials for Catalytic Applications	3	0	3
	MST F335	Coating and thin film technology	3	0	3
	MST F336	Glass Technology	3	0	3
	MST F337	Materials for Energy Applications	3	0	3
	MST F338	Metals and Alloys	3	0	3
	MST F339	Polymer Materials	3	0	3
	PHY F379	Thin Film Technology	3	0	3
	PHY F414	Physics of Advanced Materials	3	1	4
	PHY F416	Soft condensed Matter Physics	3	1	4

This minor has 3 core courses and 16 electives from which a minimum of 15 units must be chosen, and it is offered by the department of Chemical Engineering.

Materials Science and Engineering is an interdisciplinary field that utilizes the knowledge of Physics, Chemistry, Engineering, Mathematics, Biology, and Biotechnology, but has its own unique character.

It is constantly evolving- new and exciting materials, such as nanomaterials, high-temperature, and high-pressure materials, are being developed, including lightweight materials, sustainable

biomaterials, and green materials for tissue engineering. The Material Science field combines a broad knowledge base and diverse practical and commercial applications.

11. Minor in Philosophy, Economics and Politics

Courses & Units Required	6 courses (min) 18 units (min)				
Core Courses	Course Number	Course Title	L	P	U
	ECON F211	Principles of Economics	3	0	3
	GS F211	Modern Political Concepts	3	0	3
	HSS F235	Introductory Philosophy	3	0	3
Elective Courses	BITS F 385	Introduction to Gender Studies	3	0	3
	GS F231	Dynamics of Social Change	3	0	3
	GS F234	Development Economics	3	0	3
	GS F243	Current Affairs	3	0	3
	GS F312	Applied Philosophy	3	0	3
	GS F313	Marxian Thoughts	3	0	3
	GS F332	Contemporary India	3	0	3
	GS F333	Public Administration	3	0	3
	HSS F236	Symbolic Logic	3	0	3
	HSS F315	Society, Business, and Politics	3	0	3
	HSS F322	Social and Political Ecology	3	0	3
	HSS F331	Sankara's Thoughts	3	0	3
	HSS F333	Comparative Religion	3	0	3
	HSS F343	Professional Ethics	3	0	3
	HSS F345	Gandhian Thoughts	3	0	3
	HSS F346	International Relations	3	0	3
	HSS F350	Human Rights: History, Theory and Practice	3	0	3
	HSS F353	Philosophy of Aesthetics	3	0	3
	HSS F354	Introduction to Islamic Economy	3	0	3
	HSS F355	Dictatorship, Democracy & Development	3	0	3
	HSS F356	Social Movements and Protest Politics	3	0	3

With 3 core courses and 21 electives, this minor requires a minimum of 18 units for successful completion.

The minor in Philosophy, Economics, and Politics (PEP) is designed to introduce students to a variety of approaches to comprehend the social and human world in which we live and to develop skills applicable to a variety of career opportunities in national and international settings such as worldwide organizations.

It would pique the interest and enthusiasm of those students wishing to supplement their fundamental knowledge in science and

engineering possessing a solid foundation in the humanities and social sciences.

As a multi-disciplinary entity, this option will provide a balanced mix of social science knowledge, (economics, sociology, and politics) and the humanities (philosophy) that would complement the social sciences and enable students to draw political, economic, and social connections as well as equipping them with the skills to think critically.

12. Minor in Physics

Courses & Units Required	5 courses (min) 15 units (min)				
Core Courses	Course Number	Course Title	L	P	U
	PHY F212 or ECE F212/ EEE F212/ INSTR F212	Electromagnetic Theory – 1 or Electromagnetic Theory	3	0	3
	PHY F242	Quantum Mechanics – 1	3	0	3
	PHY F312	Statistical Mechanics	3	0	3
Elective Courses	BITS F316	Nonlinear Dynamics and Chaos	3	0	3
	BITS F386	Quantum Information and Computing	3	0	3
	PHY F211	Classical Mechanics	3	1	4
	PHY F213	Optics	3	0	3
	PHY F214	Electricity Magnetism and Optics Lab	0	2	2
	PHY F215	Introduction to Astronomy and Astrophysics	3	0	3
	PHY F241	Electromagnetic Theory – 2	3	1	4
	PHY F243	Mathematical Method of Physics	3	0	3
	PHY F244	Modern Physics Lab	0	2	2
	PHY F311	Quantum Mechanics – 2	3	0	3
	PHY F313	Computational Physics	3	0	3
	PHY F315	Theory of Relativity	3	0	3
	PHY F341	Solid State Physics	3	0	3
	PHY F342	Atomic and Molecular Physics	3	0	3
	PHY F343	Nuclear and Particle Physics	3	0	3
	PHY F346	Laser Science and Technology	3	0	3
	PHY F418	Lasers and Applications	3	1	4
	PHY F426	Physics of Semiconductors Devices	3	1	4
	PHY F427	Atmospheric Physics	3	0	3
	PHY F428	Quantum Information Theory	3	0	3

While the minor is not a lucrative choice from the placement perspective, it is beneficial for research purposes or higher studies. Compared to other minors (such as data science), grading and coursework are lighter in the physics minor. Though the cut-off criterion is not rigid, it is not recommended to take the physics minor for the purpose of an additional degree.

For those pursuing a single degree, choosing a minor would limit the OPEL (Open elective) options, and the minor courses would be

surplus CDCs (Compulsory disciplinary courses) which may prove as an additional burden on top of the already existing coursework.

It has been observed that single degree ECE/EEE/ENI students take up EMT (Electromagnetic theory) in 2-1, so that they have one less CDC in the Physics minor.

There have also been cases where students opted to switch careers after 2-3 years of graduating and pursue an MS abroad. Here, they were benefited as the physics minor degree aided in enhancing their college applications.

The following excerpts are insights from BITSians who had taken this minor during their respective tenures:

A 4th year student at BITS Pilani Goa Campus, who is now conducting his Thesis at Moscow Institute of Physics and Technology mentioned that minors are a great way to explore other fields outside the major, adding that courses in the Physics minor (Statistical Mechanics, Quantum Mechanics) are effort-demanding and are not easy so it is advisable to take up a Physics minor programme only if there is genuine interest.

The minor degree helps one to stand out (it proves to recruiters that one was able to handle the extra workload of completing a minor programme) during placements. However, it does not compensate for your (if) poor performance in your core discipline.

A good CGPA in your major is a must. He added that students pursuing minors get priority registration for the courses they have to take so it is easier for someone to get a course that fills up very fast if he/she is pursuing a minor with the course as a requirement.

A 4th year CS undergraduate student at BITS Pilani Goa Campus mentioned that EMT and Statistical mechanics are offered in odd

semesters and Quantum mechanics 1 is offered in even semesters. Statistical mechanics has Quantum Mechanics - 1 as a prerequisite.

So, students should take care while planning their minor. (advisable to take statistical mechanics in 2-2). If one fails to take up QM - 1 in 2-2 they would have to complete stat mech in 4-1 and cannot cover their ps-1 in 4-1 if the minor degree is to be completed.

A lot of electives also have prerequisites so one must choose electives accordingly. The AUGSD website has a doc which mentions all the prerequisites for a course.

A 4th Year student of BITS Pilani Goa Campus, pursuing a major in Physics mentioned that for students who would want to know about undergrad physics, they could choose electives that cover more breadth of the physics curriculum, i.e., courses that are introductory in nature like Nonlinear dynamics and chaos, Quantum Informatics, Computational Physics, and Theory of relativity.

Students who want to pursue some sort of research in the future should opt for courses such that they get in-depth knowledge about at least one field. For example, QM-1 + QM-2 + EMT-2 + Atomic Physics taken together forms a very good bundle to understand how atomic level physics works.

Courses closely related to the Electronic branch in the physics minor are Semiconductor devices, Nonlinear dynamics and Chaos (a kind of subset of the control system course).

He added that some popular research internships do not have prerequisites, i.e., one need not be a major in physics to get them but still, a minimum amount of coursework would be required which can be completed if the minor courses are chosen judiciously.

A degree in a physics minor proves that the person is good at working with mathematical models and if required can work

comfortably in analytical roles also. So, it may prove beneficial while interviewing.

13. Minor in Public Policy

Courses & Units Required	5 courses (min) 15 units (min)				
Core Courses	Course Number	Course Title	L	P	U
	GS F233	Public Policy	3	0	3
	GS F333	Public Administration	3	0	3
Elective Courses	HSS F232	Introduction to Development Studies	3	0	3
	HSS F317	Introduction to Globalisation	3	0	3
	HSS F322	Social and Political Ecology	3	0	3
	HSS F361	Urban Policy and Governance	3	0	3
	HSS F362	Local Governance and Participation	3	0	3
	HSS F363	Disaster and Development	3	0	3

This minor consists of 2 core courses and 5 electives, requiring a minimum selection of 15 units.

The purpose of this course is to provide students with a clear and contextualized understanding of the conceptual and empirical aspects of public policy, the nature of public policy interventions in India, and their varying impacts.

In addition, it intends to provide students with an understanding of the dynamics of policy making, as well as its central components of governance and fundamental characteristics and functions of institutions, equipping them with the skills of policy evaluation.

14. Minor in Robotics and Automation

Courses & Units Required	05 courses (min) 15 units (min)				
Core Courses	Course Number	Course Title	L	P	U
	BITS F441	Robotics	3	0	3
	EEE/INSTR/ECE F242	Control Systems	3	0	3
	BITS F327	Artificial Intelligence for Robotics	2	1	3
Electives	BITS F312	Neural Network & Fuzzy Logic	3	0	3
	BITS F415	Introduction To MEMS	3	1	4
	BITS F442	Remote Sensing and Image Processing	3	0	3
	BITS F464	Machine Learning	3	0	3
	ECE F434	Digital Signal Processing	3	1	4
	EEE F411	Internet of Things (IoT)	3	1	4
	EEE F422	Modern Control Systems	3	0	3
	EEE G512	Embedded System Design	3	1	4
	INSTR F343	Industrial Instrumentation and Control	3	0	3
	INSTR G611	Advanced Control Systems	3	2	5
	ME F221 or MF F221	Mechanisms and Machines	3	0	3
	ME F432	Computer Aided Manufacturing	2	1	3
	MF F311	Mechatronics & Automation	2	1	3
	MSE G511	Mechatronics	3	2	5

Comprising 14 electives and 3 core courses, this minor requires 15 units for completion and as the name suggests, it is a broad set of elective courses covering application of the fundamentals of robotics and automation to various industry sectors.

15. Minor in Supply Chain Analytics

Courses & Units Required	05 courses (min) 15 units (min)				
Core Courses	Course Number	Course Title	L	P	U
	BITS F455	Analytics for Supply Chain	3	0	3
	MF F319	Supply Chain Management	3	0	3
	MF F422	Supply Chain Modelling and Empirical Analysis	3	1	4
Electives	ME F443	Quality Control Assurance and Reliability	3	0	3
	MF F321	Procurement Management	3	0	3
	MF F418	Lean Manufacturing	3	0	3
	MF F485	Sustainable Manufacturing	3	0	3
	MATH F212 OR	Optimization	3	0	3
	ME F320 OR	Engineering Optimization			
	MF F320	Engineering Optimization			
	MATH F242	Operations Research	3	0	3
	MATH F353	Statistical Inference and Applications	3	0	3

Supply Chain Analytics contains 7 electives and 3 core courses, and the main objective is to assist organizations in making better, timelier, and more informed business decisions.

By 2025, the global market for supply chain analytics is expected to surpass \$10 billion, with a compound annual growth rate of 17% and 16% annual growth rate (CAGR).

Existing supply chain analytics solutions have impressive capabilities, which will only increase with future advancements.

The programme will enable students to build a solid foundation and broaden their horizons in the knowledge base of supply chain and supply chain analytics in general.

It will include supply chain management, supply chain modelling and empirical analysis.

The minor programme is intended to produce professionals in the supply chain for the current and future business environment.

16. Minor in Water and Sanitation

Courses & Units Required	05 courses (min) 15 units (min)				
Core Courses	Course Number	Course Title	L	P	U
	BIO F216	Water Sanitation and Solid Waste Management	3	0	3
	BIO F217	Laboratory for Water Sanitation and Solid Waste management	1	2	3
Electives	BIO F266	Study Project			3
	SAN G511	Sanitation Technology	3	2	5
	SAN G512	Sanitation and Public Health	3	2	5
	SAN G513	Sanitation Governance Behaviour change and Advocacy			5*
	SAN G514	Sanitation Finance and Project Management			5*
	SAN G515	Emergency Sanitation & Leadership			5*

Comprising 2 core courses and 6 electives, this minor requires 15 units for completion. The focus of Sustainable Development Goal 6 (SDG 6) is Water and Sanitation, as well as the SDG 6 tasks.

As evidenced by the Swachh Bharat Mission, the Indian government also prioritizes sanitation. Postgraduate trained and working professionals are high in demand, notable instances include The Bill and Melinda Gates Foundation, investing heavily in Water. They had funded UNESCO IHE and its eight partners through their Sanitation and Hygiene programme in the development of the e-learning alliance.

The foundation's sanitation investment strategy requires skilled and knowledgeable professionals. This minor would enable students to acquire the skills and knowledge that will enable them to gain insights in the area of water and hydrology.

Interview with Mr. AP Singh regarding Water and Sanitation:

The Students' Academic Cell reached out to Mr. AP Singh for his insights on the minor of Water and Sanitation.

Introduced to the curriculum 2 years ago, Water and Sanitation is offered every year. Sir explained that the main objective of the minor is to make water an easily available and accessible commodity to everyone, for all purposes including irrigation, agriculture, industrial as well as recreational.

He added that the minor involves water treatment processes, transportation, distribution and water treatment technologies.

By studying this minor, students would be able to learn more about the finance of projects, policy implementation and the software utilized in Water and Sanitation.

Mr. AP Singh also mentioned that the aim was to discover the ways in which society can be helped and added that all minors must be chosen according to the students' interests, and must be planned well in advance.